House Price Predictor Using Linear Regression

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Dataset: California Housing Dataset (Scikit-learn)

This project builds a Linear Regression to predict house price using the California Housing dataset provided by the Scikit-learn.

Dataset Description:

The California Housing dataset contains information about housing districts in California. The dataset has 20,640 samples and 8 numerical features describing housing characteristics such as media income, house age, average rooms and location coordinates.

Exploratory Data Analysis:

Exploratory Data Analysis was performed to understand the dataset. Summary statistics and visualizations were used to analyze distributions and correlations.

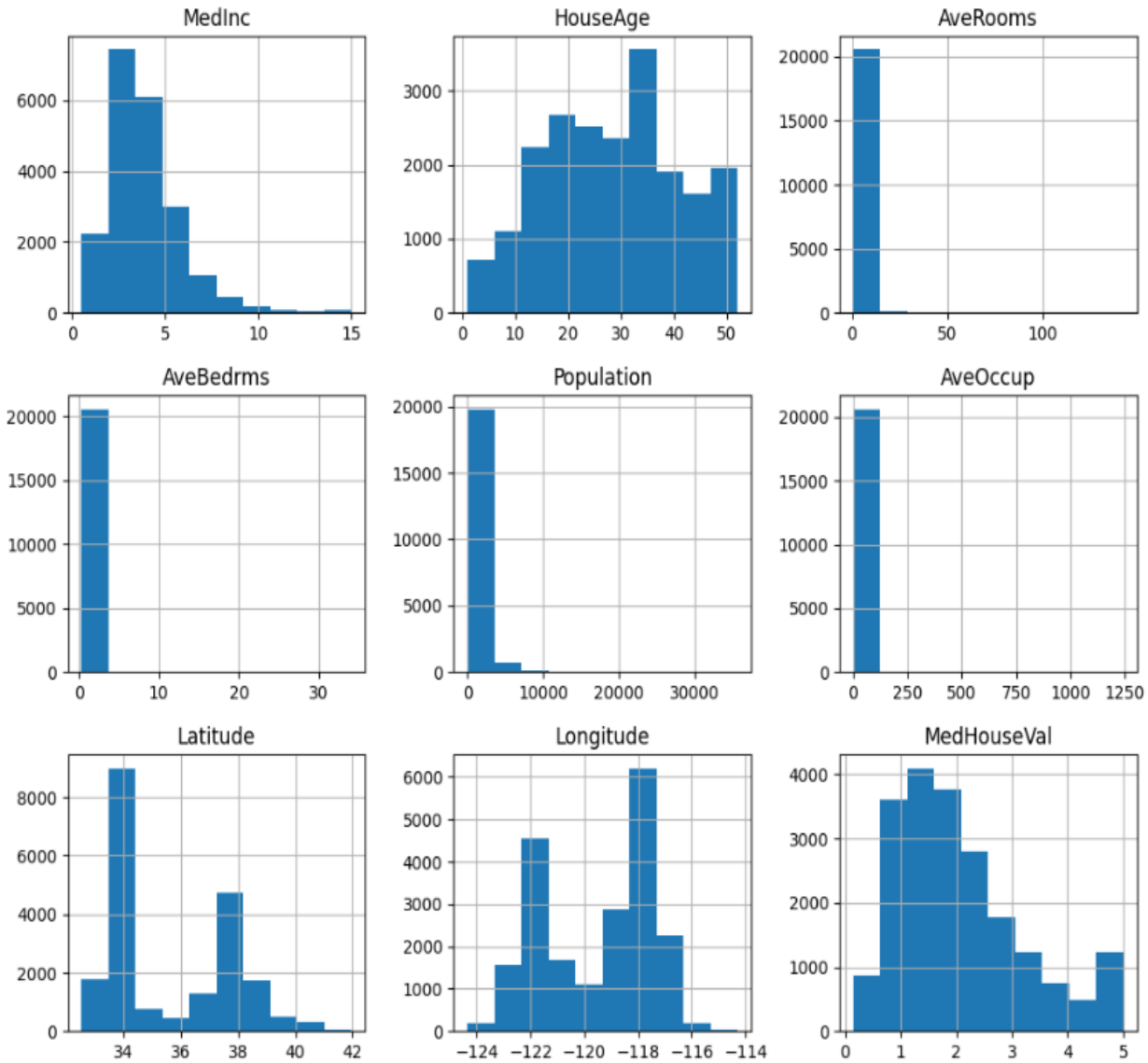
Key Observations

No missing values in the dataset.

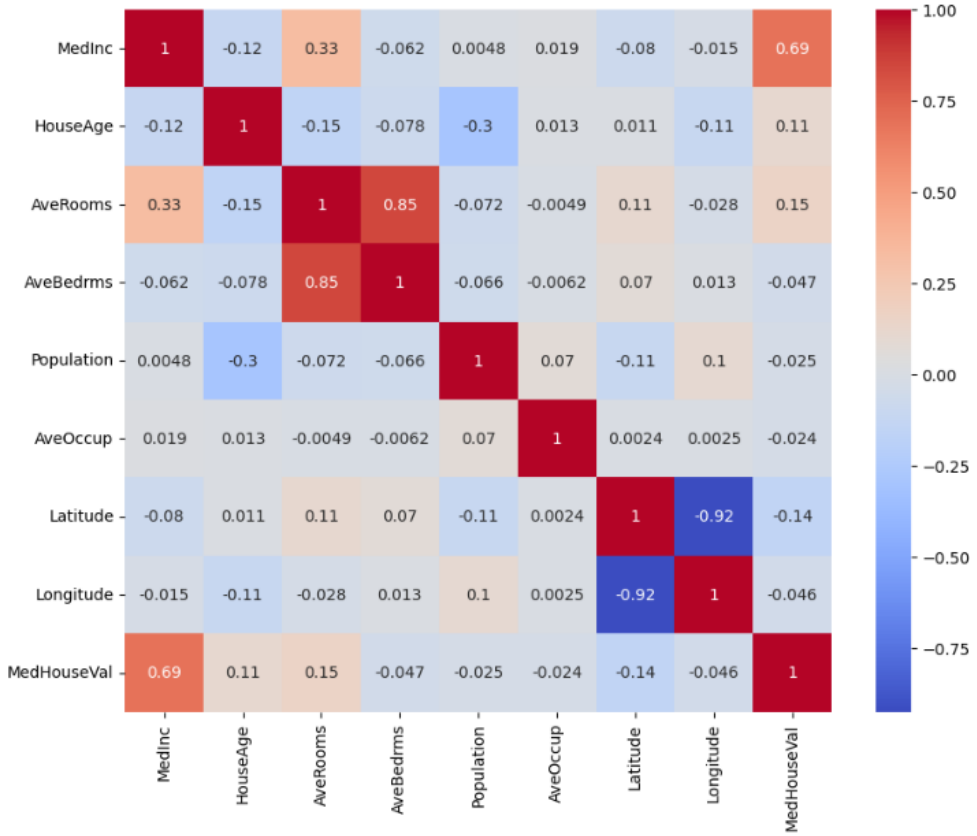
Median income has the strongest correlation with the house price

Geographic features (Latitude and Longitude) and influence prices.

Histogram Plot



Correlation heatmap



Model Training:

A linear regression model was used to predict house prices. The dataset was split into 80% training data and 20% testing data using train_test_split.

Steps performed:

1. Dataset loaded using fetch_california_housing
2. Features and target separated
3. Data split into training and testing sets
4. Linear Regression model trained using Scikit-learn

Model Evaluation:

MAE: 0.53

RMSE: 0.74

R^2 Score: 0.57